

DESIGN THINKING

UNIT II

IDEATION STAGE (IDEATE STAGE)

WHAT is the Ideate mode

Ideate is the mode of the design process in which you concentrate on idea generation.

Mentally it represents a process of “going wide” in terms of concepts and outcomes.

Ideation provides both the fuel and also the source material for building prototypes and getting innovative solutions into the hands of your users.

WHY ideate

You ideate in order to transition from identifying problems to creating solutions for your users.

Ideation is your chance to combine the understanding you have of the problem space and people you are designing for with your imagination to generate solution concepts.

Particularly early in a design project, ideation is about pushing for a widest possible range of ideas from which you can select, not simply finding a single, best solution.

The determination of the best solution will be discovered later, through user testing and feedback.

Various forms of ideation are leveraged to:

- Step beyond obvious solutions and thus increase the innovation potential of your solution set
- Harness the collective perspectives and strengths of your teams
- Uncover unexpected areas of exploration
- Create fluency (volume) and flexibility (variety) in your innovation options

HOW to ideate

You ideate by combining your conscious and unconscious mind, and rational thoughts with imagination.

For example, in a **brainstorm** you leverage the synergy of the group to reach new ideas by building on others' ideas.

Adding constraints, surrounding yourself with inspiring related materials, and embracing misunderstanding all allow you to reach further than you could by simply thinking about a problem.

Another ideation technique is building – that is, prototyping itself can be an ideation technique. In physically making something you come to points where decisions need to be made; this encourages new ideas to come forward.

There are other ideation techniques such as bodystorming, mindmapping, and sketching. But one theme throughout all of them is deferring judgment – that is, separating the generation of ideas from the evaluation of ideas. In doing so, you give your imagination and creativity a voice, while placating your rational side in knowing that you will get to the examination of merits later.

Brainstorming

What is Brainstorming?

Finding a solution to a particular problem is frequently achieved through brainstorming, a collaborative creative practice. This is done by allowing team members to contribute new ideas, which are gathered and recorded freely. A director or facilitator oversees brainstorming sessions, which many core team members attend.

It's possible to come up with many ideas during a brainstorming session that you wouldn't have access to by simply sitting down with a pen and paper. By interacting with one another, listening to one another, and expanding on one another's ideas, brainstorming aims to maximize the group's collective thinking.

Stages Of Brainstorming

The brainstorming process typically involves three stages: preparation, ideation, and evaluation.

Preparation: The focus of the preparation stage should be on setting the rules, structure, and culture around the session. It may, for example, encourage team members to come to the team with sufficient background knowledge, and ensure all people in the group know the importance of creating a non-judgemental environment.

Ideation: The ideation stage involves sharing ideas which will be added to the brainstorming notes, such as on a flipboard or shared screen during a group video call . This is where the actual idea generation takes place. Participants are encouraged to share their ideas knowing that there is 'no silly idea' at this point in time.

Evaluation: With a wide range of ideas collected, the group needs to categorize, review, and select the most promising ideas. This may involve drawing connections between ideas, merging ideas together, and finding identifying problems with certain ideas. At this stage, it's important to ensure the chosen ideas have alignment with the objective.

Importance of Brainstorming in Design Thinking

Business brainstorming in design thinking can aid you in getting your creativity flowing, whether you're trying to develop successful marketing strategies, commercials, or slogans.

Diverse Viewpoints

Any organization that relies on a single person to come up with all of its ideas will eventually run out of fresh ideas or start to repeat themselves. But at a brainstorming session, you might get ideas from many people. These ideas might not be amazing or workable, but when you brainstorm with others, they might develop into something new and useful.

Help Organise Ideas

Brainstorming aids in planning and idea organization. By brainstorming, you can develop a sound plan for what to write and what to leave out. For instance, it is beneficial for teachers to prepare earlier and create lesson plans for their pupils in advance.

After you've planned, brainstorming also aids in keeping your ideas and thoughts organized. Write down the first thing that comes to mind. Afterward, organize it by selecting the top concept from all the good ideas and filtering them.

Strengthens Team Working

Instead of placing undue pressure on one person to come up with a fresh, original idea, brainstorming enables all team members to attempt to share their thoughts. This promotes the development of team relationships and encourages team members to treat one another with respect and be accountable for their job.

Unique Ideas

You will be able to generate a lot of original ideas through brainstorming. These concepts all exhibit originality in the same way that philosophers and innovators devised novel ideas by adopting brainstorming techniques. You don't need to worry about coming up with original ideas because everything will be natural and organic.

Rules for Brainstorming

- Only one conversation is allowed at a time. No other person must intervene when an idea is being given.
- Focus must be on the quantity and not on quality. In this step, the group must have large

number of ideas with them.

- Think out of the blue. Wild ideas must be encouraged even if they invoke plain humor or seem impossible.
- The group leader must defer judgment. The fellow thinkers also need to suspend judgment. Judgmental attitude leads to an obstruction for the thinkers.
- Visualization is important. The design thinkers must create a visual picture of the problem statement and then try to see a visual image of their ideas as well.
- Build on each other's ideas. Support other ideas and build on them through group discussions and healthy debates.
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10 Examples Of Brainstorming

Below are some possible situations in which brainstorming can be highly effective.

Example 1: Product Development

Brainstorming can be used to generate new product ideas or improve upon existing ones. For example, a team of designers, engineers, and marketers could brainstorm ideas for a new smartphone that incorporates cutting-edge technology and features. Importantly, the team should be composed of product market experts and, ideally, people with prior knowledge about issues with the current product iteration, consumer feedback, and gaps in the marketplace.

Example 2: Marketing Campaigns

Brainstorming is common in marketing and advertising, and in fact, the term was coined by a marketing professional.

Generally, this session would involve bringing together a team of creatives with good knowledge of the market as well as cutting-edge marketing techniques in order to come up with a campaign idea. For example, a team of marketers could brainstorm ideas for a new social media campaign that leverages the power of influencers to reach the audience.

Example 3: Brainstorming For A Novel

Brainstorming can be an excellent approach to improve writing techniques, especially when writing a novel.

In this situation, I would gather some fellow writers or personal tutors who have experience developing plots, characters and themes and go over the things that could work better in your novel.

By analyzing my plot structure and understanding my character's traits based on their backstory, I could get valuable insight into how to make the story more engaging.

Example 4: Brainstorming For Business Strategic Plans

Brainstorming is an excellent way to devise strategic plans for higher-level business development. It helps you visualize how your business may look like in the future while allowing feedback from team members involved in the development process to obtain insights from all departments.

A team of executives may get together around a single table with reports and data sheets explaining different growth areas of the company.

Example 5: Brainstorming New Classroom Ideas

Brainstorming is one of the best ways for teachers to develop new ideas for curriculum building and lesson planning.

Teachers should consider mingling with colleagues who have years teaching experience engaging students around different scenarios shaping them towards positive cognitive outcomes.

Example 6: Brainstorming Home Decor Projects

When renovating a home, brainstorming can help skyrocket creativity while considering factors like budget, style, and functionality.

Collaborating with an interior designer or friends who have taste in home decor and DIY projects can be useful in generating interesting ideas that match the requirements of the homeowner.

Example 7: Brainstorming For Event Planning

Brainstorming is an essential tool when it comes to event planning as it helps to identify key themes, vendors, catering, and decoration ideas.

The best part of brainstorming is involving event planners together with their clients in a room or a virtual hangout session to discuss their vision for the occasion and generate ideas in real-time.

Example 8: Brainstorming Personal Life Goals

Brainstorming can help you set achievable personal goals while shedding light on your desires. At times like these having a life coach might come in handy who can incorporate exercises where you jot down all the things you desire either professionally or personally such as traveling to other countries or buying a new house.

Example 9: Brainstorming UX Designs

In the development of digital products such as web applications or mobile apps brainstorming plays a key role.

Through group discussions between UX designers and developers they'll emphasize ways of

enhancing user experience by identifying areas where previous iterations had no success.

Example 10: Brainstorming Career Choices

Brainstorming can help young people finishing high school to create a roadmap towards the best career for them.

At this time of life, people usually don't have a clear idea of the job they will do, but they may have a clear idea of what they are good at, what they enjoy doing, and the general direction they want to go (white collar, blue collar, etc.).

The process of deciding what to do may involve seeking out mentors or attending career fairs where people can offer guidance and support.

Benefits Of Brainstorming

The benefits of brainstorming are numerous. It can help people and organizations generate new ideas, solve complex problems, and make better decisions.

In the workplace, it can also improve team morale and strengthen team cohesion. By engaging individuals in idea generation, companies can create a culture of innovation and creativity.

1. Innovation

Firstly, brainstorming plays a significant role in boosting innovation.

When we sit together and come up with different creative ideas, we tend to approach situations with new perspectives that we often overlook alone. Sometimes our minds can only go so far when left to its devices!

The act of bouncing thoughts off one another elevates creativity tremendously. Brainstorming as a group often produces new solutions that wouldn't have surfaced otherwise .

2. Problem Solving

Secondly, brainstorming is incredibly beneficial for problem-solving.

While we all face challenges in life, brainstorming can act as a beneficial tool for addressing and overcoming those issues.

When faced with a problem, having multiple people collaborate during the decision-making process leads to better outcomes than relying solely on one person's point of view.

In addition, when each member contributes equally unique views and suggestions about possible solutions without dismissing others' input or ideas, new strategies can arise which become successful approaches .

3. Team Morale And Cohesion

A side-effect of brainstorming as a group is that it can make a stronger group dynamic. Its key principles include inclusion, open-mindedness, and working together.

Coincidentally, this can also make work much more enjoyable!

Collaborating as a team creates cohesiveness within the company culture because all persons contribute towards achieving mutual goals rather than accomplishing solo achievements only related to their title or job description .

Support from team members through both triumphs and failures can increase mutual respect among colleagues for each other while creating social bonds.

4. Culture Of Innovation

Lastly, creating a culture of innovation becomes achievable when utilizing brainstorming tasks regularly within the company environment.

Brainstorming can lead to creative solutions that would not be possible without the open-minded, free-flowing brainstorming process .

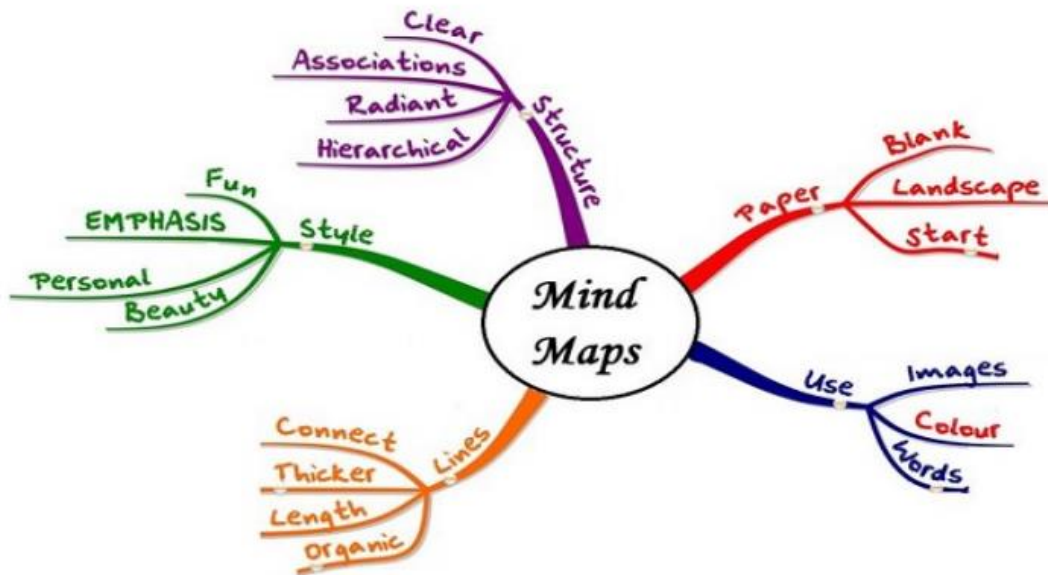
Mind Maps

- Mind map is a diagram that helps to observe and study information in a visual manner.
- Mind map is created around a single problem statement and all the ideas to solve the problem are written around it.
- The problem statement usually is written at the center of a blank page as a hub and branches shoot out in all directions representing the solutions.
- The ideas can be represented as text, images, trees, and even smaller mind maps.
- The entire map looks like a top view of a tree, with the problem statement as the trunk and the solutions as branches.
- It is also known by the name of spider diagram.
- However, mind map is not a mere haphazard diagram. It is a well-structured organized diagram meant to aid the thinking process and to streamline the analysis and synthesis process.

Guidelines to Create Mind Maps

- Begin with the problem statement at the center of a blank white page.
- Use images, different colors, symbols, caricatures, abbreviations and codes to depict your ideas.
- Text can be boring, but different depictions can add an altogether different charm to your mind map.
- Keywords must replace long statements.
- The mind map must give a hint to the design thinker about an idea quickly.

- Reading a long statement is waste of time.
- Each and every word written in the mind map Figure(shown below) must be connected to the central hub by some or other line or set of lines. Use multiple colors for visual stimulation. \Use radial hierarchy and make use of emphasis, italics, and underlines to stress on a point.

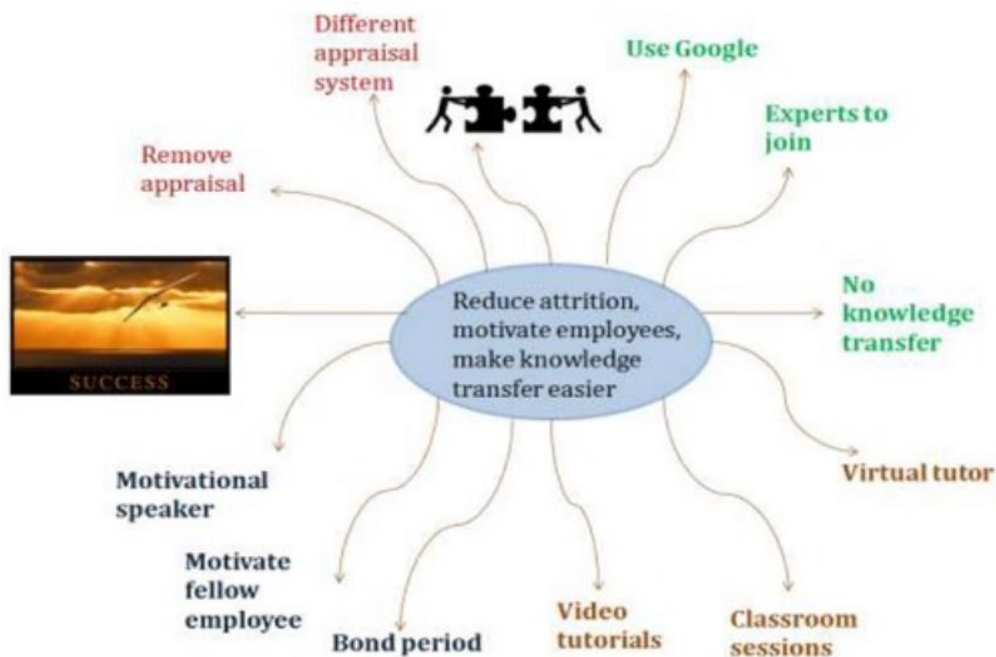


Mind Maps: for example the case of schools.

The knowledge transfer program is nothing much different than schools teaching the students.

How does a school manage to keep the students motivated towards studies?

How does a school teach the kids?



**Question Bank:
(Short Answers)**

Q.No.	Question	Reference page number (Above)for answer
1	Explain in brief Ideate mode?	1
2	Why to ideate in design Thinking?	1
3	List four methods of Ideation?	1,2
4	Explain the term “Brainstorming”?	2
5	List the three stages of Brainstorming?	2
6	List four benefits of Brainstorming?	6,7

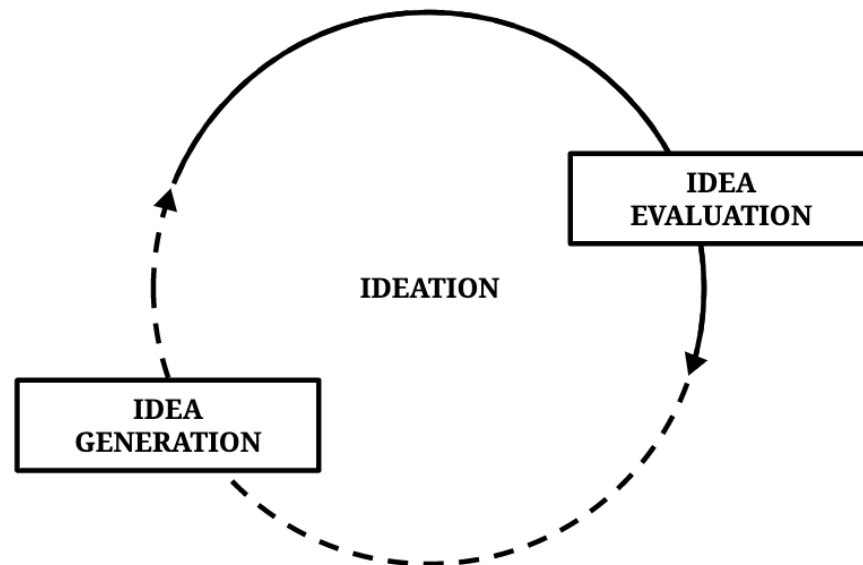
**Question Bank:
(Descriptive Answers)**

Q.No.	Question	Reference page number (Above)for answer
1	Apply brainstorming to any two areas of application?	4,5,6
2	Explain the importance of Brainstorming?	3
3	List the rules necessary for effective Brainstorming?	3,4
4	List guidelines for crating mind maps?	7,8

UNIT II(PART B – 50%)

Idea evaluation: the other side of the creativity

Ideation is often thought to be mostly about idea generation. However, the process of developing ideas doesn't stop there: it's followed by an evaluation phase. Balancing idea generation with idea evaluation is crucial to the creative process.



“Within the field of creativity, idea generation in general and the usefulness of idea generation methods has received much greater attention than idea evaluation,” write Gerard Puccio and John Cabra in the *Handbook of Organizational Creativity*.

This lack of research may be due to the mysticism surrounding ideation. In Greek mythology, the muses served as intermediaries on behalf of the gods, who provided inspiration for creative ideas.

Today, we don't quite believe in muses anymore, but we still look at ideas as this magical phenomenon which cannot be explained, and cannot be controlled. In the words of Carl Jung: “Any reaction to stimulus may be causally explained; but the creative act, which is the absolute antithesis of mere reaction, will forever elude human understanding. It can only be described in its manifestation; it can be obscurely sensed, but never wholly grasped.”

But our perception of the creative process doesn't have to feel so obscure. While ideation is fascinating indeed, it's not a black box we may never get to fully understand.

Ideation is an iterative process (2 key elements of Ideation)

Instead of a specific Eureka moment, researchers have identified two main elements in the ideation process, which involves the “generation of candidate ideas or solutions followed by extensive exploration of those ideas.”

Idea generation. This phase of the creative process is about coming up with many blind variations to solve a problem. Quantity is the measure of performance.

Idea evaluation. In this phase, selective retention is applied by assessing how practical, useful or relevant each idea is. The goal is to only keep quality ideas.

“Idea generation could be thought of as a search for novelty, while its process partner, idea evaluation, might be thought of as an effort to make novel thinking practical, useful or relevant.”

— Gerard Puccio and John Cabra.

Steps involved in Idea Evaluation

Define your evaluation criteria. Before you start evaluating your ideas, consider what inclusion and exclusion criteria are relevant to your current project. An idea may be exciting but out-of-budget, or you may have some specific time constraints. List all of these factors so you have an objective way to assess your raw ideas.

Get the data you need. Do not rely on intuition only to evaluate your ideas. Especially for complex problems, make sure to get all the information you need. Read the latest research, consult with experts, and seek advice from your colleagues.

Consider using an evaluation framework. Beyond the pass-fail method, there are many evaluation frameworks you may find helpful in refining a range of ideas. A popular one is the Strengths, Weaknesses, Opportunities and Threats (SWOT) analysis, but you can create your own evaluation matrix.

Allocate sufficient time for the evaluation process. Maybe brainstorming sessions dedicate 90% of the time to generating ideas, and rush at the end to choose a short selection to keep on exploring. Make sure you have enough time for the second phase of ideation.

Follow up over the long term. The evaluation process should be continuous. Monitor the project to ensure the ideas you selected are providing the solution you expected. If that’s not the case, try to understand why: did the problem change? Was the idea not adequate in the first place? Document what you learn so it can be used for future projects.

Generating lots of ideas but failing to select the most appropriate ones is unlikely to result in positive creative outcomes. By making sure you value the evaluation process as much as the idea generation process, you are more likely to develop better creative ideas.

6 Creativity Techniques to Foster Creative Thinking

Numerous techniques have been devised to foster creative thinking. Below are descriptions of some of these techniques:

1. **Attribute Listing:** This technique was developed by Robert Crawford. It involves listing all attributes of an object or problem and then attempting to modify each attribute or group of attributes in as many ways as possible. The aim is to generate novel ideas and explore different possibilities.
2. **Brainstorming:** Developed by Alex P. Osborn, this technique encourages a group of people to freely and uninhibitedly suggest solutions to a problem. Participants are allowed to generate as many alternatives as possible without criticism, leading to the emergence of creative ideas.
3. **Gordon Technique (Synectics):** Gordon Technique, or Synectics, is a creative method by William J. Gordon. It's a variant of brainstorming, breaking limitations and encouraging a broader outlook. Participants are kept in the dark about the problem, promoting diverse ideas. Group has varied backgrounds, leader facilitates, allowing unrestricted thinking for unique solutions. Success depends on a leader's understanding. Draws inspiration from literary works for unconventional problem-solving with imagination. Empowers creative minds for novel solutions.
4. **Seek Outside Help:** Though not explicitly described in detail, this technique suggests seeking external perspectives or input from individuals outside the immediate group to gain fresh insights and ideas.
5. **Nominal Group Technique:** The Nominal Group Technique is a method where the group functions in name only. The process involves these steps: the leader introduces the problem to the group; (b) each member independently writes down their ideas; (c) each member shares their best idea, which is displayed on a blackboard for all to see; (d) a discussion is held to explain and evaluate the ideas. The main goal is to limit interpersonal communication and maximize contributions from members. The technique follows a highly structured approach to encourage creative thinking from each individual.
6. **Delphi Technique:** This technique involves collecting information from physically dispersed individuals through a written questionnaire. The responses are collected and summarized in a feedback report, allowing for anonymous and unbiased input, particularly useful when dealing with complex and distributed problem-solving scenarios.

4th Stage / phase of Design Thinking - Prototype phase

WHAT is the Prototype mode/stage/phase?

The Prototype mode is the iterative generation of artifacts intended to answer questions that get you closer to your final solution. In the early stages of a project that question may be broad – such as “do my users enjoy cooking in a competitive manner?” In these early stages, you should create low-resolution prototypes that are quick and cheap to make (think minutes and cents) but can elicit useful feedback from users and colleagues. In later stages both your prototype and question may get a little more refined. For example, you may create a later stage prototype for the cooking project that aims to find out: “do my users enjoy cooking with voice commands or visual commands”.

A prototype can be anything that a user can interact with – be it a wall of post-it notes, a gadget you put together, a role-playing activity, or even a storyboard. Ideally you bias toward something a user can experience. Walking someone through a scenario with a storyboard is good, but having them role-play through a physical environment that you have created will likely bring out more emotions and responses from that person.

WHY prototype?

1. **To ideate and problem-solve.** Build to think.
2. **To communicate.** If a picture is worth a thousand words, a prototype is worth a thousand pictures.
3. **To start a conversation.** Your interactions with users are often richer when centered around a conversation piece. A prototype is an opportunity to have another, directed conversation with a user.
4. **To fail quickly and cheaply.** Committing as few resources as possible to each idea means less time and money invested up front.
5. **To test possibilities.** Staying low-res allows you to pursue many different ideas without committing to a direction too early on.
6. **To manage the solution-building process.** Identifying a variable also encourages you to break a large problem down into smaller, testable chunks.

HOW to prototype?

1. **Start building.** Even if you aren't sure what you're doing, the act of picking up some materials (post-its, tape, and found objects are a good way to start!) will be enough to get you going.
2. **Don't spend too long on one prototype.** Let go before you find yourself getting too emotionally attached to any one prototype.
3. **Identify a variable.** Identify what's being tested with each prototype. A prototype should answer a particular question when tested. That said, don't be

blind to the other tangential understanding you can gain as someone responds to a prototype.

4. **Build with the user in mind.** What do you hope to test with the user? What sorts of behavior do you expect? Answering these questions will help focus your prototyping and help you receive meaningful feedback in the testing phase.

Concept Prototype

Embodying concepts in tangible forms to get feedback from users

WHAT IT DOES

A Concept Prototype is used to assess the adoptability of an early concept among teams and with potential users by giving it physical form that can be experienced. Concept Prototypes embody the principle of “building to learn” in that the process of giving a physical form to a concept facilitates discoveries about it that often cannot be foreseen until it is made tangible. What emerges from the method is a kind of “reality check” that helps teams make well-informed decisions about the direction a concept needs to evolve.”

HOW IT WORKS

STEP 1: Identify concepts to be prototyped.

Review concepts to identify those that will benefit most from testing in tangible form. Determine the kinds of readily available materials you will need to create a rough embodiment of the concept. Determine what kind of prototype—appearance prototype, performance prototype, or a combination of both—will be most useful at that stage of the process.

STEP 2: Create a prototyping space and build to learn.

It is a good idea to identify or create a space where your team can build and test Concept Prototypes. Assemble the kinds of materials and tools you will use for building, modifying, and testing prototypes.

STEP 3: Review prototypes, test, and discuss.

Present Concept Prototypes to your team in a review session and to a group of users. Test these prototypes and discuss each one in light of the stated design principles, user needs, form factors, human factors, and other dimensions. How does it feel to hold the prototype? Is it intuitive? Comfortable? Does it make sense?

STEP 4: Modify prototypes and iterate.

Build on existing prototypes, modify them, or create new ones to reflect input gathered through the review process. Iterate and continue to incorporate feedback into the prototype.

STEP 5: Summarize key learning.

Use the conclusion of the review sessions to record the key learning and results from testing. Summarize how the prototype evolved from an initial manifestation to a final desired state. Share this information among team members and stakeholders to reinforce decisions about further development.

BENEFITS

Encourages iterations

Makes abstract ideas concrete Promotes collaboration

INPUT

Concepts that can benefit from testing in tangible form

OUTPUT

Refined concepts adjusted around how potential users interact with prototypes

Question Bank: (Short Answers)

Q.No.	Question	Reference page number (Above)for answer
1	Explain the term “idea generation”?	11
2	Explain the term “idea evaluation”?	11
3	List first 3 techniques out of 6 Creativity Techniques to Foster Creative Thinking?	12
4	List last 3 techniques out of 6 Creativity Techniques to Foster Creative Thinking?	12
5	What is the Prototype mode/stage/phase?	13
6	Give two benefits of concept prototype?	15
7	What is the output of concept prototype?	15
8	What is the input to concept prototype?	15

**Question Bank:
(Descriptive Answers)**

Q.No.	Question	Reference page number (Above)for answer
1	Identify & explain the steps involved in Idea Evaluation?	11
2	Identify & explain 6 Creativity Techniques to Foster Creative Thinking?	12
3	Identify & explain the reasons for “why to prototype”?	13
4	Identify & explain for “how to prototype”?	13,14
5	Identify & explain how concept prototype works?	14,15